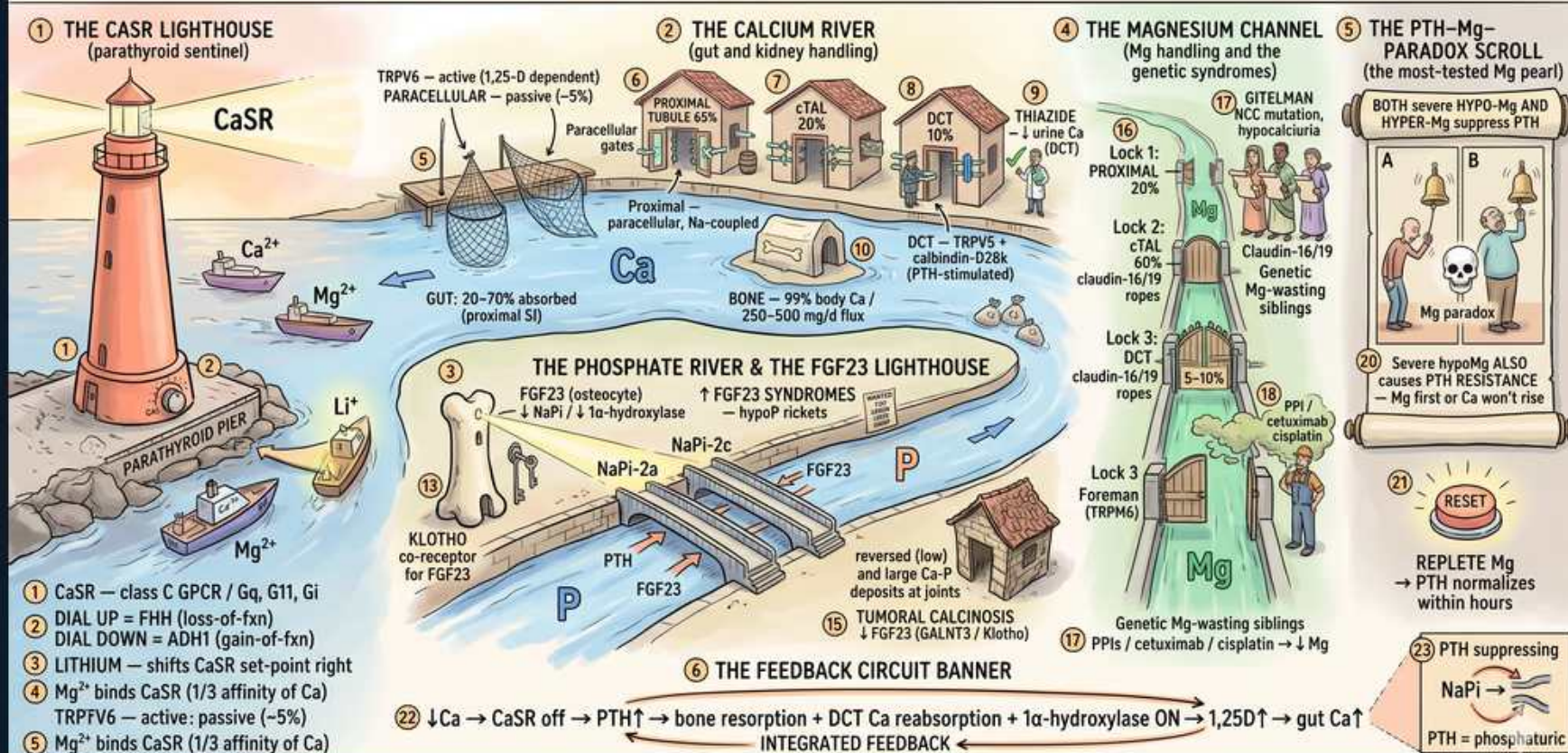


The Three-River Harbor — Calcium, Phosphate & Magnesium Trafficking

THE THREE-RIVER HARBOR — CALCIUM, PHOSPHATE & MAGNESIUM TRAFFICKING

Normal Ca 8.5–10.5 mg/dL (50% ionized) — P 2.5–4.5 mg/dL — Mg 1.7–2.4 mg/dL — corrected Ca = +0.8 per 1.0 g/dL low albumin



Bottom banner: Mg first when hypocalcemia is refractory. CaSR is the master sensor. Thiazides ↓ urine Ca (DCT); loops ↑ urine Ca (cTAL). FGF23 = phosphate's ↓1,25-D enforcer.

1. Normal calcium values

WHAT YOU SEE

Banner: Ca 8.5-10.5, ionized 50%

WHAT IT ENCODES

Total Ca normal 8.5-10.5 mg/dL with ~50% ionized; phosphate 2.5-4.5; Mg 1.7-2.4. Correct total Ca for albumin: add 0.8 mg/dL per 1.0 g/dL drop in albumin. Ionized calcium is the physiologically active fraction.

BOARD TRAP

Wrong answer: low total Ca in hypoalbuminemia = true hypocalcemia. Discriminator: correct for albumin; ionized Ca may be normal.

2. CaSR — the master sensor (GPCR)

WHAT YOU SEE

CaSR lighthouse, parathyroid sentinel

WHAT IT ENCODES

The calcium-sensing receptor is a Gq/G11 GPCR on parathyroid and thick ascending limb. High Ca activates CaSR → suppresses PTH and increases urinary Ca. It is the upstream master regulator of calcium homeostasis.

BOARD TRAP

Wrong answer: PTH is the primary Ca sensor. Discriminator: CaSR senses calcium; PTH is the downstream effector hormone.

3. CaSR loss-of-function → FHH

WHAT YOU SEE

Dial UP = loss of function, FHH

WHAT IT ENCODES

Inactivating CaSR mutation (dial up set-point) causes Familial Hypocalciuric Hypercalcemia: mild hypercalcemia, inappropriately normal/high PTH, but LOW urine calcium (Ca/Cr clearance ratio <0.01). Benign, avoid parathyroidectomy.

BOARD TRAP

Wrong answer: primary hyperparathyroidism (do surgery). Discriminator: FHH has LOW urinary Ca (FECa <1%); 24h urine Ca distinguishes it — do NOT operate.

4. CaSR gain-of-function → ADH

WHAT YOU SEE

Dial DOWN = gain of function, ADH

WHAT IT ENCODES

Activating CaSR mutation (dial down set-point) causes Autosomal Dominant Hypocalcemia: hypocalcemia with inappropriately low PTH and HIGH urine Ca. Treating with calcium worsens hypercalciuria/nephrocalcinosis.

BOARD TRAP

Wrong answer: hypoparathyroidism, give aggressive Ca. Discriminator: ADH has hypercalciuria — over-replacement causes nephrocalcinosis; treat cautiously.

5. Lithium — shifts CaSR set-point

WHAT YOU SEE

Lithium label at lighthouse

WHAT IT ENCODES

Lithium shifts the CaSR set-point, causing a biochemical picture resembling FHH/primary HPT: mild hypercalcemia with high-normal PTH and reduced urinary Ca. Resolves with lithium withdrawal in many cases.

BOARD TRAP

Wrong answer: primary hyperparathyroidism needing surgery. Discriminator: drug history of lithium mimics FHH; check the medication list first.

6. Calcium river — gut & kidney handling

WHAT YOU SEE

Calcium river, PCT and DCT

WHAT IT ENCODES

Gut absorbs ~20-70% of Ca (1,25-vitamin D dependent). Renally, ~99% is reabsorbed: bulk paracellularly in PCT/thick ascending limb, fine-tuned actively in DCT via TRPV5 (PTH-stimulated). Bone is the large reservoir.

BOARD TRAP

Wrong answer: most renal Ca handling is in the DCT. Discriminator: bulk reabsorption is proximal/TAL; DCT only fine-tunes the final ~5-10%.

7. Thiazide → DCT Ca reabsorption

WHAT YOU SEE

Thiazide arrow at DCT

WHAT IT ENCODES

Thiazides act at the DCT and increase calcium reabsorption (decrease urine Ca) — useful in calcium-stone formers and hypercalciuria. Loops do the opposite. This is a heavily tested diuretic-calcium pairing.

BOARD TRAP

Wrong answer: thiazides cause hypercalciuria. Discriminator: thiazides LOWER urine Ca (raise serum Ca); loops RAISE urine Ca.

8. Phosphate river & FGF23

WHAT YOU SEE

Phosphate river, NaPi-2a, FGF23

WHAT IT ENCODES

Phosphate is reabsorbed proximally via NaPi-2a cotransporters. Both PTH and FGF23 are phosphaturic (internalize NaPi-2a → phosphate wasting). FGF23 also suppresses 1,25-vitamin D. Klotho is the FGF23 co-receptor.

BOARD TRAP

Wrong answer: PTH retains phosphate. Discriminator: PTH is PHOSPHATURIC (wastes phosphate) while raising calcium.

9. FGF23 syndromes — osteomalacia/rickets

WHAT YOU SEE

FGF23 syndromes box

WHAT IT ENCODES

Excess FGF23 (X-linked hypophosphatemic rickets, tumor-induced osteomalacia, autosomal dominant hypophosphatemic rickets) causes renal phosphate wasting, low phosphate, inappropriately low/normal 1,25-D, and osteomalacia.

BOARD TRAP

Wrong answer: vitamin D-deficient rickets. Discriminator: FGF23 excess gives LOW phosphate with low 1,25-D despite normal calcium; treat with phosphate + calcitriol/burosumab.

10. Klotho co-receptor for FGF23

WHAT YOU SEE

Klotho label at phosphate lighthouse

WHAT IT ENCODES

Klotho is the obligate co-receptor that confers FGF23 specificity in the kidney. Without Klotho, FGF23 cannot signal. CKD features Klotho deficiency and very high FGF23 levels.

BOARD TRAP

Wrong answer: FGF23 acts alone. Discriminator: FGF23 requires Klotho co-receptor for its phosphaturic and vitamin D-suppressing actions.

11. Magnesium channel — TRPM6, claudins

WHAT YOU SEE

Magnesium channel, TAL and DCT

WHAT IT ENCODES

Mg is reabsorbed paracellularly in the TAL (claudin-16/19) and actively in the DCT via TRPM6. Gitelman and genetic Mg-wasting syndromes (claudin-16/19 mutations) cause hypomagnesemia with hypercalciuria.

BOARD TRAP

Wrong answer: Mg handling is mainly proximal. Discriminator: most regulated Mg reabsorption is in the TAL (paracellular) and DCT (TRPM6).

12. Drugs causing Mg wasting

WHAT YOU SEE

PPI, cisplatin, aminoglycoside ropes

WHAT IT ENCODES

PPIs (chronic), cisplatin, aminoglycosides, amphotericin, and calcineurin inhibitors cause renal magnesium wasting. Hypomagnesemia impairs PTH secretion and action, producing refractory hypocalcemia.

BOARD TRAP

Wrong answer: treat the hypocalcemia with calcium alone. Discriminator: low Mg causes functional hypoPTH — REplete magnesium first or hypocalcemia persists.

13. Mg-first paradox — replete Mg first

WHAT YOU SEE

PTH-Mg paradox scroll, RESET

WHAT IT ENCODES

Severe hypomagnesemia both suppresses PTH secretion AND causes PTH resistance, so hypocalcemia is refractory. Bottom-line board rule: in hypocalcemia, REplete MAGNESIUM FIRST before expecting calcium to correct.

BOARD TRAP

Wrong answer: keep giving IV calcium for refractory hypocalcemia. Discriminator: check and correct Mg first — Mg-dependent hypocalcemia won't respond to Ca alone.

14. Feedback circuit summary

WHAT YOU SEE

Feedback circuit banner

WHAT IT ENCODES

Integrated loop: low Ca → CaSR off → PTH rises → bone resorption + DCT Ca reabsorption + 1-alpha-hydroxylase → 1,25-D → gut Ca absorption; PTH also wastes phosphate. FGF23 and phosphate enforce balance.

BOARD TRAP

Wrong answer: PTH raises both Ca and phosphate. Discriminator: PTH raises Ca but LOWERS phosphate (phosphaturic); 1,25-D raises both.